

## 4.2 Power Series Worksheet

1. Determine the Taylor series centered at 9 for  $f(x) = e^{-2x}$ .

We want to find:  $\sum_{n=0}^{\infty} \frac{f^{(n)}(9)}{n!} (x-9)^n$  which means we need to determine  $f^{(n)}(9)$ .

We note that:

$$f(x) = e^{-2x} \rightarrow f(9) = e^{-18}$$

$$f'(x) = -2e^{-2x} \rightarrow f'(9) = -2e^{-18}$$

$$f''(x) = 4e^{-2x} \rightarrow f''(9) = 4e^{-18}$$

$$f'''(x) = -8e^{-2x} \rightarrow f'''(9) = -8e^{-18}$$

...

So we see that  $f^{(n)}(x) = (-2)^n e^{-2x}$ , so  $f^{(n)}(9) = (-2)^n e^{-18}$ . Therefore, the Taylor series is:

$$\sum_{n=0}^{\infty} \frac{(-2)^n e^{-18}}{n!} (x-9)^n = \sum_{n=0}^{\infty} \frac{(-2)^n}{n! e^{18}} (x-9)^n$$

2. Determine the Taylor series centered at 1 for  $f(x) = x^3$ .

We want to find:  $\sum_{n=0}^{\infty} \frac{f^{(n)}(1)}{n!} (x-1)^n$  which means we need to determine  $f^{(n)}(1)$ .

We note that:

$$f(x) = x^3 \rightarrow f(1) = 1$$

$$f'(x) = 3x^2 \rightarrow f'(1) = 3$$

$$f''(x) = 6x \rightarrow f''(1) = 6$$

$$f'''(x) = 6 \rightarrow f'''(1) = 6$$

$$f^{(k)}(x) = 0 \rightarrow f^{(k)}(1) = 0 \text{ for all } k \geq 4$$

Therefore, we see that since  $f(x)$  is already a polynomial the Taylor series actually has finitely many terms and can be written as:

$$\begin{aligned} \sum_{n=0}^{\infty} \frac{f^{(n)}(1)}{n!} (x-1)^n &= \frac{1}{0!} (x-1)^0 + \frac{3}{1!} (x-1)^1 + \frac{6}{2!} (x-1)^2 + \frac{6}{3!} (x-1)^3 \\ &= 1 + 3(x-1) + 3(x-1)^2 + (x-1)^3 \end{aligned}$$

3. Determine the Maclaurin series centered at 0 for  $f(x) = \cos(2x)$

We want to find:  $\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} (x-0)^n$  which means we need to determine  $f^{(n)}(0)$ .

We note that:

$$f(x) = \cos(2x) \rightarrow f(0) = 1$$

$$f'(x) = -2\sin(2x) \rightarrow f'(0) = 0$$

$$f''(x) = -4\cos(2x) \rightarrow f''(0) = -4$$

$$f'''(x) = 8\sin(2x) \rightarrow f'''(0) = 0$$

$$f^{(4)}(x) = 16\cos(2x) \rightarrow f^{(4)}(0) = 16$$

...

We can see that in general that every odd derivative will evaluate to be 0 and every even derivative will evaluate to be a power of 4 (with alternating sign).

Therefore, the Maclaurin series is:

$$\sum_{n=0}^{\infty} \frac{(-4)^n}{(2n)!} (x-0)^{2n} = \sum_{n=0}^{\infty} \frac{(-4)^n x^{2n}}{(2n)!}$$

4. Determine the interval and radius of convergence for the series  $\sum_{n=1}^{\infty} n(x+5)^{n+1}$ .

We use the Ratio Test

$$\begin{aligned} \lim_{n \rightarrow \infty} \left| \frac{(n+1)(x+5)^{(n+1)+1}}{n(x+5)^{n+1}} \right| &= \lim_{n \rightarrow \infty} \frac{(n+1)|x+5|^{n+2}}{n|x+5|^{n+1}} = \lim_{n \rightarrow \infty} \frac{(n+1)}{n} |x+5| \\ &= \lim_{n \rightarrow \infty} \left( \frac{n+1}{n} \cdot \frac{1/n}{1/n} \right) |x+5| = \lim_{n \rightarrow \infty} \left( \frac{1+1/n}{1} \right) |x+5| = |x+5| \end{aligned}$$

So, by the Ratio Test we know that this series is convergent when

$|x+5| < 1 \Leftrightarrow -1 < x+5 < 1 \Leftrightarrow -6 < x < -4$ , and it is divergent when  $|x+5| > 1$ . The test is inconclusive when  $x = -6$  and  $x = -4$ . So, we will analyze these cases separately.

- When  $x = -6$  the series becomes  $\sum_{n=1}^{\infty} n(-6+5)^{n+1} = \sum_{n=1}^{\infty} n(-1)^{n+1}$ . We note that  $\lim_{n \rightarrow \infty} [n(-1)^{n+1}]$  DNE since the odd terms increase towards infinity and the even terms decrease towards negative infinity. Therefore, by the Test for Divergence the series diverges.
- When  $x = -4$  the series becomes  $\sum_{n=1}^{\infty} n(-4+5)^{n+1} = \sum_{n=1}^{\infty} n(1)^{n+1} = \sum_{n=1}^{\infty} n$ , which we know diverges by the Test for Divergence.

Therefore we have an interval of convergence of  $-6 < x < -4$  and radius of convergence of  $R = 1$ .

5. Find the interval and radius of convergence for the power series  $\sum_{k=0}^{\infty} \left(\frac{9}{4}\right)^k x^{2k}$ .

We use the Ratio Test: 
$$\lim_{k \rightarrow \infty} \left| \frac{\left(\frac{9}{4}\right)^{k+1} x^{2(k+1)}}{\left(\frac{9}{4}\right)^k x^{2k}} \right| = \lim_{k \rightarrow \infty} \frac{9}{4} \cdot \frac{|x|^{2k+2}}{|x|^{2k}} = \lim_{k \rightarrow \infty} \frac{9}{4} |x|^2 = \frac{9}{4} |x|^2 = \frac{9}{4} x^2.$$

So, by the Ratio Test for Absolute Convergence we know that this series is convergent when  $\frac{9}{4} x^2 < 1 \Leftrightarrow x^2 < \frac{4}{9} \Leftrightarrow |x| < \frac{2}{3} \Leftrightarrow -\frac{2}{3} < x < \frac{2}{3}$ , and it is divergent when  $|x| > \frac{2}{3}$ . The test is inconclusive when  $x = -\frac{2}{3}$  and  $x = \frac{2}{3}$ . So, we will analyze these cases separately.

- When  $x = -\frac{2}{3}$  or  $x = \frac{2}{3}$  the series becomes  $\sum_{k=0}^{\infty} \left(\frac{9}{4}\right)^k \left(\pm \frac{2}{3}\right)^{2k} = \sum_{k=0}^{\infty} \left(\frac{9}{4}\right)^k \left(\frac{4}{9}\right)^k = \sum_{k=0}^{\infty} (1)^k = \sum_{k=0}^{\infty} 1$ . We note by the Test for Divergence that this series must diverge.

Therefore we have an interval of convergence of  $-\frac{2}{3} < x < \frac{2}{3}$  and radius of convergence of  $R = \frac{2}{3}$ .

6. Find the interval and radius of convergence for the power series  $\sum_{k=1}^{\infty} \frac{(x-2)^k}{k}$

We use the Ratio Test:

$$\lim_{k \rightarrow \infty} \left| \frac{\frac{(x-2)^{k+1}}{k+1}}{\frac{(x-2)^k}{k}} \right| = \lim_{k \rightarrow \infty} \frac{|x-2|^{k+1}}{k+1} \cdot \frac{k}{|x-2|^k} = \lim_{k \rightarrow \infty} \left( \frac{k}{k+1} \right) |x-2| = \lim_{k \rightarrow \infty} \left( \frac{k}{k+1} \cdot \frac{1/k}{1/k} \right) |x-2| = \lim_{k \rightarrow \infty} \left( \frac{1}{1/k+1} \right) |x-2| = |x-2|$$

So, by the Ratio Test we know that this series is convergent when  $|x-2| < 1 \Leftrightarrow -1 < x-2 < 1 \Leftrightarrow 1 < x < 3$ , and it is divergent when  $|x-2| > 1$ . The test is inconclusive when  $x=1$  and  $x=3$ . So, we will analyze these cases separately.

- When  $x=1$  the series becomes  $\sum_{k=1}^{\infty} \frac{(1-2)^k}{k} = \sum_{k=1}^{\infty} \frac{(-1)^k}{k}$ . This is alternating harmonic series which we know converges by the Alternating Series Test.
- When  $x=3$  the series becomes  $\sum_{k=1}^{\infty} \frac{(3-2)^k}{k} = \sum_{k=1}^{\infty} \frac{(1)^k}{k} = \sum_{k=1}^{\infty} \frac{1}{k}$ . This is the harmonic series which we know diverges.

Therefore we have an interval of convergence of  $1 \leq x < 3$  and radius of convergence of  $R=1$ .

7. Find the interval and radius of convergence for the power series  $1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \frac{x^6}{6!} + \frac{x^8}{8!} + \dots$

We need to start by writing this series using summation notation. We note that  $\sum_{k=0}^{\infty} \frac{x^{2k}}{(2k)!}$

Now we use the Ratio Test: 
$$\lim_{k \rightarrow \infty} \left| \frac{\frac{x^{2(k+1)}}{(2(k+1))!}}{\frac{x^{2k}}{(2k)!}} \right| = \lim_{k \rightarrow \infty} \frac{|x|^{2k+2}}{(2k+2)!} \cdot \frac{(2k)!}{|x|^{2k}} = \lim_{k \rightarrow \infty} \frac{|x|^2}{(2k+2)(2k+1)} = 0.$$

So, by the Ratio Test we know that this series is convergent when  $0 < 1$ , and since this is true for all values of  $x$  we have an interval of convergence of  $(-\infty, \infty)$  and radius of convergence of  $R = \infty$ .

*Note:* This series is actually the Maclaurin series for  $f(x) = \cosh(x)$ !